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The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: Results from a pilot-scale digester treating sewage sludge

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HIGHLIGHTS

- The benefits that oxygen can provide under unbalanced conditions are studied.
- A microaerobic digester is subjected to a hydraulic overload.
- Micro-oxygenation seems to prevent a severe imbalance.
- The reactor's productivity increases during the imbalance.
- Micro-oxygenation improves the biogas quality independently of the HRT.

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ABSTRACT

A well-functioning pilot reactor treating sewage sludge at approximately 4.4 NL/m^3/d of oxygen supply and 18 d of hydraulic retention time (HRT) was subjected to a hydraulic overload to investigate whether oxygen benefits successful operation in stressful circumstances. Only a mild imbalance was caused, which was overcome without deterioration in the digestion performance. Volatile solids (VS) removal was 45% and 43% at 18 and 14 d of HRT, respectively. Biogas productivity remained around 546 NmL/g_{VS}, but it was slightly higher during the period of imbalance. Thereafter, similar performances were achieved. Under anaerobic conditions, VS removal and biogas productivity were respectively 41% and 525 NmL/g_{VS}, hydrogen partial pressure rose, and acetic acid formation became less favourable. Oxygen seemed to form a more stable digestion system, which meant increased ability to deal successfully with overloads. Additionally, it improved the biogas quality; methane concentration was negligibly lower, while hydrogen sulphide and oxygen remained around 0.02 and 0.03% v/v, respectively.

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1. Introduction

Anaerobic digestion is an established technology for the treatment of many wastes from different origins. Recent European policies encourage its application seeking environmental and socio-political targets (Pöschl et al., 2010). However, low methane yield and poor operational stability still hinder its widespread application (Chen et al., 2008).

Several authors have identified hydrolysis as the rate-limiting step in anaerobic digestion of particulate substrates, such as sewage sludge (Appels et al., 2008). Recent studies have demonstrated that it can be enhanced by introducing limited amounts of oxygen (or air) directly into the anaerobic digester (Jenicek et al., 2008) or during a pre-treatment step (Jagadabhi et al., 2010). In fact, Botheju et al. (2010a) managed to raise methane yield by micro-aeration. Nonetheless, only the hydrolysis of carbohydrates and proteins have reported improvements (Johansen and Bakke, 2006).

According to Zhu et al. (2009), the basis of the hydrolytic effect of oxygen resides in the increased synthesis and activity of extracellular hydrolytic enzymes. Similarly, Botheju and Bakke (2011) pointed to enhanced growth rates of facultative acidogens and the consequent larger release of these enzymes. Cellular growth stimulation is indeed a well-known oxygen benefit that has been exploited to raise the yield of some biochemical conversions (Chen et al., 2003).

It is an accepted issue that strict anaerobes (acetogens and methanogens) have several deterrence mechanisms to tolerate microaerobic conditions with no or minor inhibitory effects

Abbreviations: AN14, anaerobic period at 14 d of HRT; MA14, microaerobic period at 14 d of HRT; MA18, microaerobic period at 18 d of HRT; ORR, organic removal rate; PA, partial alkalinity; TA, total alkalinity.

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(Botheju and Bakke, 2011). Nevertheless, the presence of oxygen can induce other negative effects on methane potential, such as excessive oxidation of readily available substrates by fermentative (acidogenic) biomass (Johansen and Bakke, 2006) or methane consumption by aerobic methanotrophs (Zitomer and Shrout, 1998); hence the need for micro-oxygenation optimisation. Zhu et al. (2009) showed that the efficiency of the hydrolysis depends on the micro-aeration level. On the other hand, oxygen can cause positive effects on the digesters' productivity by alleviating sulphide inhibition (Zhou et al., 2006). Furthermore, under certain conditions, simultaneous sulphide removal from both the gas and liquid phase is feasible (Díaz et al., 2010b). Therefore, microaerobic conditions can also be applied to enhance biogas quality.

Obviously, the specific operational conditions that have been applied in every research have determined the overall oxygen impact on digestion; hence the different results reported. Nguyen et al. (2007) achieved higher methane yield by micro-aeration without evidence of improvement in hydrolysis. Conversely, Díaz et al. (2010a) effectively desulphurised biogas from a sewage sludge digester without affecting substrate conversion or methane yield, which is consistent with Ye et al. (2005). They found that, while pre-aeration of the biofilm carriers of three anaerobic attached film expanded reactors did not lead to improved digestion performance under normal circumstances, its resistance and recovery speed from hydraulic and organic overloads, as well as start-up rapidity, rose in comparison with the anaerobic ones. Moreover, they reported significantly lower VFA concentrations in the effluents from the pre-aerated reactors. This is consistent with Botheju et al. (2010b), who pointed out that oxygen could help digesters to confront shock loads by VFA oxidation, and the consequent prevention of pH instabilities. In fact, Zitomer and Shrout (1998) observed faster recovery from a pH drop under microaerobic conditions.

The main objective of this study was to recognise and describe the possible benefits that oxygen can provide under stressful circumstances. Therefore, the performance and stability of a pilot sludge digester treating sewage sludge at non-standard (short) hydraulic retention time (HRT) were evaluated under both microaerobic and anaerobic conditions.

2. Methods

2.1. Pilot-scale digester

Digestion was carried out in a continuous stirred tank reactor (CSTR) of 250 L total volume and 50 L headspace. It was heated at 35 °C by an electric resistor; temperature was maintained by insulating the walls of the digester. Mixed sludge with a variable organic load was collected weekly from a wastewater treatment plant, and stored at 4 °C. It was pumped to the bioreactor from a continuously stirred tank at ambient temperature. HRT ranged from 14 to 18 d depending on the operational stage (Table 1). As a result, organic loading rate (OLR) fluctuated between 1.4 and 2.9 $kg_{VS}/m^3/d$. Microaerobic conditions were implemented by supplying pure oxygen into the sludge recirculation, which was set at 50 L/h. Pure oxygen has been proved to be more profitable than air when applying microaerobic conditions (Díaz et al., 2010a).

Table 1
Operational conditions.

Period	MA18	MA14	AN14
Duration (d)	23	54	35
HRT (d)	18	14	14
OLR ($kg_{VS}/m^3/d$)	1.7	2.1	1.9
Conditions	Microaerobic	Microaerobic	Anaerobic
Oxygen flow rate ($NL/m^3/d$)	4.4	5.8–6.2	0

2.2. Monitoring and experimental analysis

Digestion pressure and temperature were monitored by using a sensor and a probe, respectively. Biogas production was measured volumetrically by water displacement, and its composition in terms of methane, carbon dioxide, nitrogen, oxygen, hydrogen sulphide and hydrogen was determined by gas chromatography (GC) according to Díaz et al. (2010b). All this data was displayed and stored in real-time in a computer.

In order to evaluate digestion performance, the feed and the digestate were analysed in terms of total and soluble chemical oxygen demand (TCOD and SCOD), total and volatile solids (TS and VS) by standard methods (APHA, 1998). Additionally, total Kjeldahl nitrogen (NKT) and ammonium were measured according to APHA (1998).

As chemical indicators of the bioreactor state, pH was monitored by a pH-meter with a temperature probe, and the alkalinity measurements (total alkalinity and partial alkalinity, TA and PA, respectively) were based on standard methods (APHA, 1998), and the methodology proposed by Ripley et al. (1986). Acetic, propionic, butyric, isobutyric, valeric, isovaleric and hexanoic acids were analysed by GC.

In order to keep track of the sulphur inputs and outputs, sulphate and thiosulphate were determined by ion chromatography and high-performance liquid chromatography (HPLC), respectively. This last method was applied according to van der Zee et al. (2007). Additionally, dissolved sulphide concentration was measured by potentiometric titration with a silver/sulphide ion selective electrode (APHA, 1998).

2.3. Experimental procedure

The digester operated under 18 d of HRT and microaerobic conditions during the 40 d preceding this study (before MA18). Therefore, data obtained in MA18 was considered the baseline of the stationary state (Table 1). Oxygen supply was interrupted from the 23rd to 25th day, to approximate more accurately the hydrogen sulphide flow rate removed in the adjacent microaerobic stages. Once it was restarted, the HRT was sharply lowered to 14 d by increasing the feeding rate (MA14). That value was set according to Díaz et al. (2011), who reported that the maximum methane production under oxygen-limiting conditions is reached within approximately 14 d. It must be noted that the micro-oxygenation was adjusted at the beginning of both MA18 and MA14 to achieve a high biogas desulphurisation performance and minimum oxygen flow rate leaving the reactor. Thus, the capacity of the digestion system to tolerate shock loads and return to a stationary state (robustness) was evaluated. Finally, the role of oxygen in the performance and stability of the process under short HRT (14 d) and anaerobic conditions was assessed (AN14).

3. Results and discussion

3.1. Digestion performance

The bioreactor adapted immediately to the shorter HRT and consequent increase in OLR (Table 1), thereby demonstrating a great hydrolytic capability. Higher organic removal rates (ORR) were reached in MA14 (0.92 against 0.75 $kg_{VS}/m^3/d$), and as a result, the biogas production rose from an average of 0.97–1.13 $Nm^3/m^3/d$ (Fig. 1a). Considering the profile of both SCOD (Fig. 1b) and total VFA concentration (Fig. 2a), higher hydrolysis rates were definitely reached.

The ORR decreased in AN14 (0.78 $kg_{VS}/m^3/d$) (Fig. 1a); hence the lower average biogas production in relation to MA14

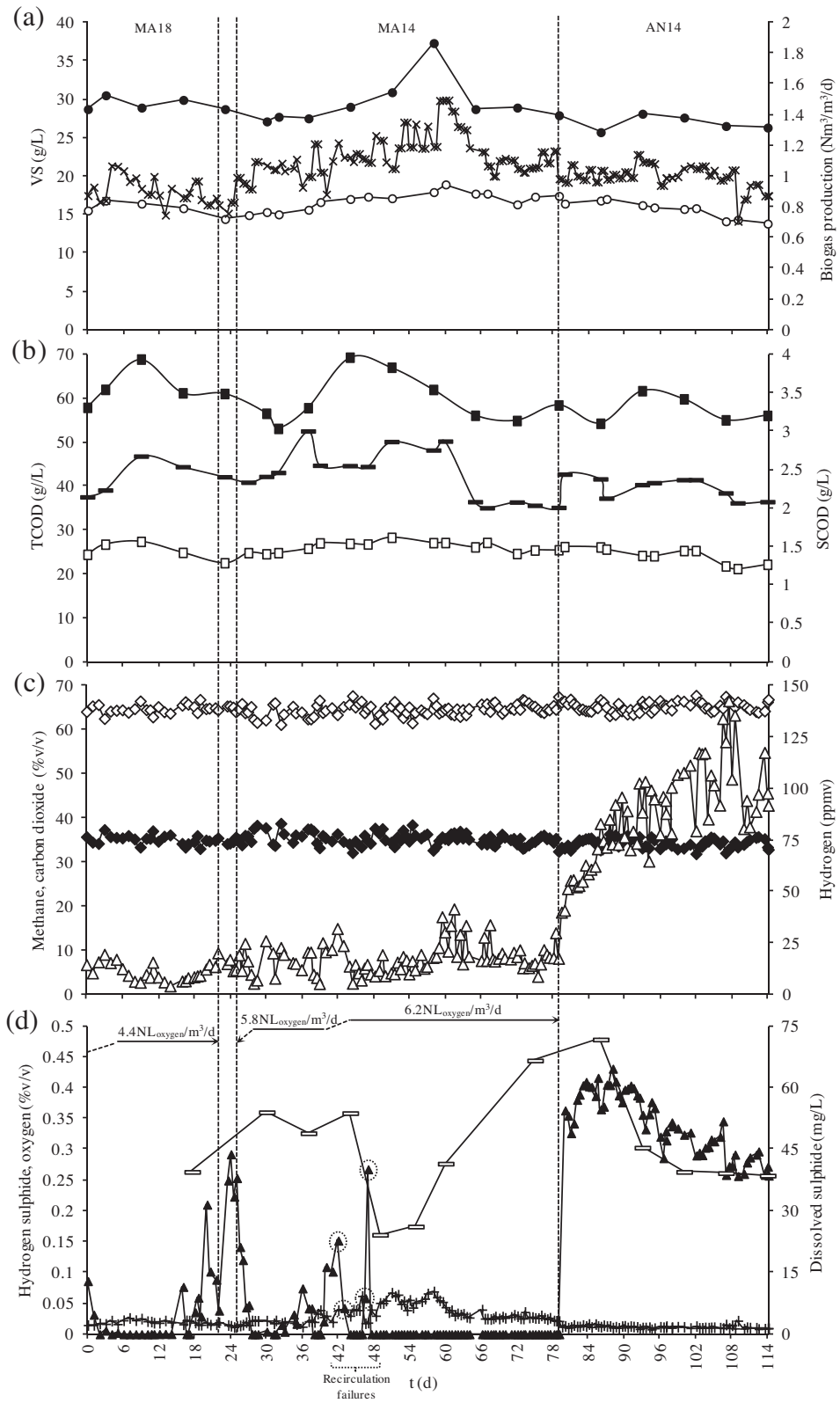


Fig. 1. (a) VS in sewage sludge (●) and in digestate (○), and biogas production (×). (b) TCOD in sewage sludge (■), and TCOD (□) and SCOD (▬) in digestate. (c) Methane (◇), carbon dioxide (⧫) and hydrogen (Δ) concentration in biogas. (d) Hydrogen sulphide (▲) and oxygen (+) concentration in biogas, dissolved sulphides in the reactor (═), and micro-oxygenation level (above).

(1.00 $Nm^3/m^3/d$). Nevertheless, it was attributed to the reduction in the OLR rather than to the absence of oxygen (Table 1). On the other hand, although this parameter remained fairly stable all through that stage, the digester's VS content was declining (Fig. 1a), which in turn explained the increasing removal of TCOD (Fig. 1b). Considering the findings of Botheju and Bakke (2011), it pointed to lower growth rates of facultative biomass under anaerobic conditions. In fact, it is possible that the increased rates in this

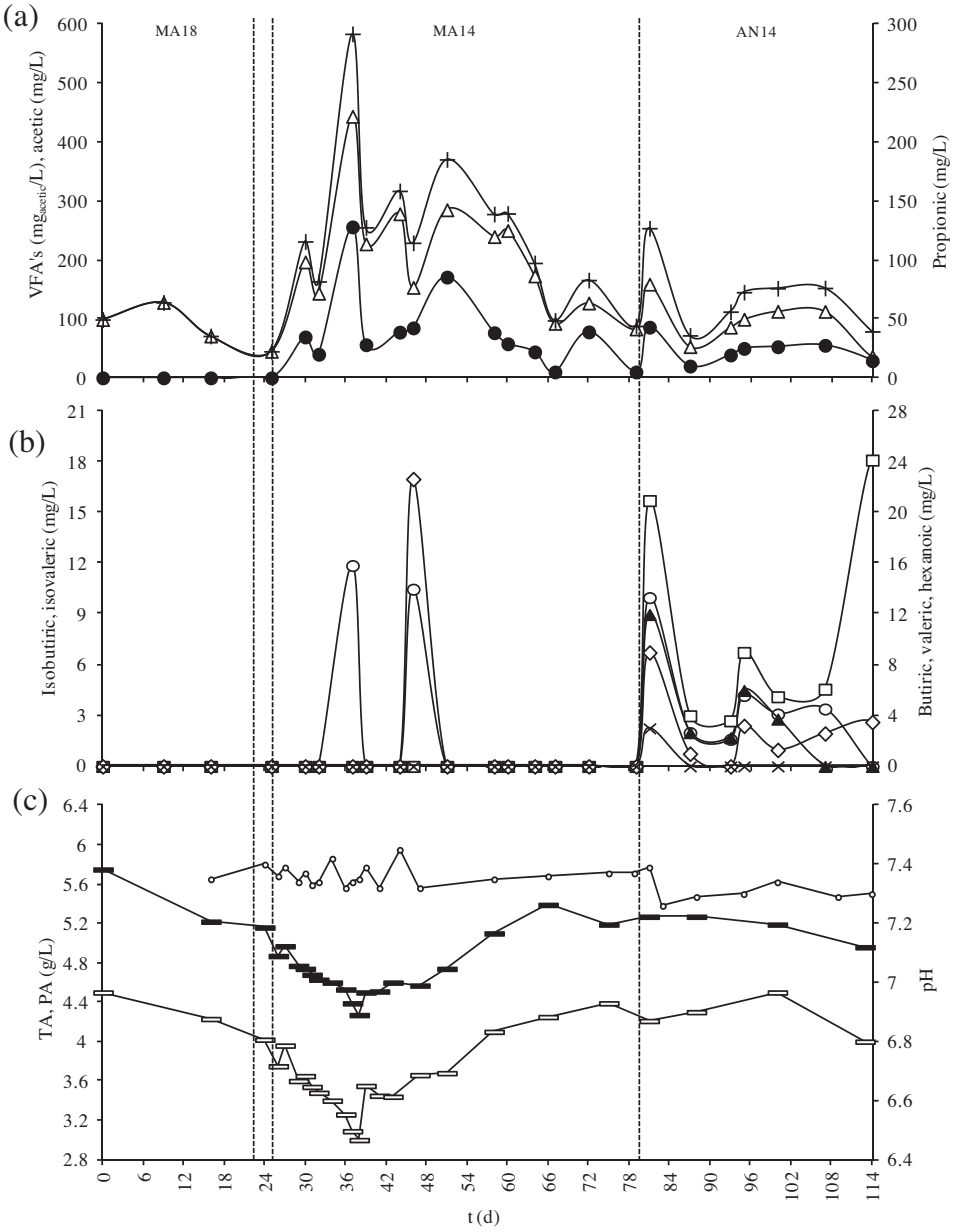


Fig. 2. (a) Total VFA concentration (+), and acetic (Δ) and propionic (●) acid. (b) Butyric (□), isobutyric ▲, valeric (◇), isovaleric (○), and hexanoic (×) acid. (c) TA (▬), PA (═), and pH (◦).

bacterial group in the presence of oxygen helped the digester to overcome the shock load without any deterioration in its productivity (see below). Furthermore, it must be taken into account that higher OLRs were confronted in MA14 (Fig. 1a).

The biogas methane content was slightly higher under anaerobic conditions, which suggested that some soluble substrate (including VFA's) could be aerobically oxidised during the microaerobic periods (Table 2). In this regard, it is noteworthy that the ratio of methane to carbon dioxide was especially variable in the first half of MA14 (Fig. 1c). This was attributed to the pH fluctuations, which regulated the solubility of the gases (Fig. 2c). Nonetheless, brief rises in aerobic activity or alkalinity consumption could also explain the momentary increases in carbon dioxide concentration, and the resulting lower biogas methane content.

On the other hand, taking into account that the highest VFA levels were reached in MA14, the lowest average of both biogas productivity and methane yield were expected at this point. However, they were obtained in AN14, while similar average values were reached in MA18 and MA14 (Table 2). Nonetheless, it must be noted that both parameters varied more during the first half of MA14 than in the rest of the stages, and they remained fairly stable from approximately the 50th day. Accordingly, it was shown that

Table 2
Digestate and biogas quality, and digestion productivity.

Period	MA18	MA14	AN14
TCOD (g/L)	25.6	26.0	24.4
SCOD (g/L)	2.5	2.4	2.2
VS (g/L)	16.1	16.7	16.0
Methane (%v/v)	64.7	64.5	65.2
Hydrogen sulphide (%v/v)	0.03	0.02	0.34
Oxygen (% v/v)	0.02	0.03	0.01
Biogas productivity (NmL/g$_{VS}$)	547	546	525
Methane yield (NmL/g$_{VS}$)	354	352	342

oxygen enhanced digestion performance only under stressful conditions, which indeed agreed with Ye et al. (2005). Similarly, previous research has demonstrated that digestion performance is not affected (neither positively nor negatively) by oxygen injection during steady-state operation at 20 d of HRT (Díaz et al., 2010a, 2010b).

3.2. Digestion stability

Acetic acid was the only VFA detected during MA18, and its concentration remained fairly stable below 128 mg_{acetic}/L (Fig. 2a). Along with the low biogas hydrogen content (11 ± 4 ppmv) (Fig. 1c), it was indicative of a well-balanced methanogenic system (Schink, 1997). Once the overload was imposed, larger amounts of both VFAs and hydrogen were released due to increased hydrolysis, acidogenesis, and acetogenesis. Almost all the extra hydrogen released was promptly consumed, which was consistent with the research pointing to the hydrogen availability being the factor limiting the hydrogenotrophic methanogens (Demirel and Scherer, 2008). Along with sulphate-reducing bacteria, they are the main hydrogen-utilising microorganisms. Nevertheless, since sulphate was provided only occasionally and in low concentrations and thiosulphate was not fed to the digester, the sulphate-reducing activity was considered to be substrate-limited (Stams et al., 2003). Accordingly, the methanogenic archaea were presumably the responsible for that result.

Hydrogen concentration in biogas started to increase instantly after the micro-oxygenation was stopped; it stabilised at approximately 100 ppmv (Fig. 1c). According to Tang et al. (2004), it could be due to the decreasing population size of hydrogen-consuming methanogens, which indeed consisted with the declining VS concentration. In their study, this microbial group proved to highly tolerate oxygen by considerably increasing its population size under microaerobic conditions. Additionally, it is worth highlighting that the lower pH maintained in AN14 could certainly have resulted from the higher hydrogen levels (Fig. 2c).

Stephenson et al. (1999) found increased hydrogen-scavenging activity under oxygen-limiting conditions, which they attributed to a higher release of growth factors to methanogens by facultative microorganisms. Accordingly, the declining concentration of acidogenic biomass in AN14 (Fig. 1a) could result in decreasing rates of hydrogenotrophic methanogenesis. Furthermore, considering that at higher hydrogen partial pressure this reaction is thermodynamically favoured (Boe, 2006), those substances definitely played a key role in stabilising the process at short HRT (see below). As shown in Fig. 1c, the hydrogen concentration remained at baseline values when the oxygen supply was interrupted at standard HRT (23rd–25th days). On the other hand, the aforementioned hydrogenotrophic effect was reported to disappear at increasing micro-oxygenation levels (Stephenson et al., 1999). Therefore, the micro-oxygenation rates set were probably crucial in achieving this effect.

According to Fig. 2a, an imbalance between the kinetics of production and the consumption of VFAs (basically propionic and acetic) took place in MA14 (Fig. 2a). The propionic acid accumulation agreed with Pind et al. (2002), who found that propionic-degraders were the acetogenic bacteria with the lowest specific growth rates. However, these microorganisms adapted surprisingly rapidly; although acetotrophic methanogens have significantly faster growth rates (Angedilaki et al., 1999), propionic was entirely degraded when acetic acid returned to the baselines. On the other hand, though in low concentrations, valeric and isovaleric were only detected when the highest ratios of propionic to acetic were reached (Fig. 2b). Hence, as Nielsen et al. (2007) indicated, this parameter was found to be a reliable indicator of the status of the process.

While VFAs accumulated, PA was gradually consumed to neutralise additional VFAs; hence the decrease in TA (Fig. 2a and c). It was indeed sufficiently high enough to avoid a pH drop. From the 37th day, the VFA consumption rates started to equal the production ones, leading to a gradual alkalinity regeneration. According to Fig. 2, the system re-stabilised from approximately the 66th day. However, only 2 d after the micro-oxygenation was stopped (around 1HRT before the process re-normalisation), the total VFA concentration doubled, and the contribution of all the VFAs increased, apart from acetic (Fig. 2a and b). Similarly, Botheju et al. (2010b) found that the higher the oxygen load, the lower the concentration of every measured VFA. Although the total VFA concentration remained close to the baseline levels thereafter, its composition continued to be more diverse. Along with acetic and propionic, butyric acid was the most important VFA during AN14. In fact, it accounted for 25% on the last day of the study, while acetic contribution dropped to 55%. Based on the above observations, it indicated the formation of a more unstable digestion system.

The shift in VFA distribution was linked to the higher content of hydrogen in the biogas. According to Boe (2006), acetogenic reactions and the fermentation pathway from simple substrates to acetic acid (that is, the formation of direct substrate to aceticlastic methanogens), carbon dioxide and hydrogen became less thermodynamically favourable in AN14. Since VFA levels re-normalised prior to the micro-oxygenation stop, its diversification did not affect digestion performance. However, the hydraulic overload in the absence of oxygen could have resulted in increasing accumulation of "other" VFAs at the expense of a reduction in acetic acid formation, which in turn could have brought on a much more severe process imbalance, or even the reactor failure. Moreover, insufficient hydrogen-scavenging activity could also have cause a pH drop. Hence, oxygen could sustain both acetoclastic and hydrogenotrophic methanogenesis in MA14, which could certainly prevent pH instabilities and help to maintain digestion performance during the imbalance.

3.3. Biogas quality

Obviously, the increased degradation rate of organic compounds (including those containing sulphur, such as proteins) maintained at 14 d of HRT led to a rise in hydrogen sulphide production; hence the higher oxygen demand in order to achieve equivalent desulphurisation efficiencies in MA14 (≈6.0 against 4.4 $NL/m^3/d$). It was indeed confirmed in AN14 (Fig. 1d).

As illustrated in Fig. 1d, the digester's sulphide content determined the hydrogen sulphide concentration in biogas according to the liquid–gas equilibrium; hence the high similarity between the profiles of these variables in AN14 (Fig. 1d). Likewise, negative correlation was found between dissolved sulphide and biogas oxygen content in MA14.

The average removal efficiency of hydrogen sulphide was estimated as 90% in MA20 and MA14 (Table 2). This percentage was consistent with the tight micro-oxygenation levels set, which were insufficient to prevent some of the concentration peaks (Fig. 1d). It should be noted that some of them (circled with a dotted line) arose from clogging problems in the recirculation stream. Hence, the biogas was successfully desulphurised independently of the HRT. Besides, considering both its negligibly lower methane concentration and the minute content of oxygen during the microaerobic periods (Table 2), it could be affirmed that the biogas quality was enhanced in relation to AN14. As a result, considering the above conclusions, benefits of micro-oxygenation on the digestion process and the biogas quality could be achieved simultaneously with oxygen supplies aiming for efficient hydrogen sulphide removal from biogas, thereby maximising the profit from micro-

oxygenation. Therefore, in full-scale, where oxygen concentrators could be employed in order to apply microaerobic conditions, the supply of limited amounts of oxygen could certainly be cost-effective.

As indicated above, most of the oxygen supplied was consumed inside the digester (90–95%) (Fig. 1d). Elemental sulphur was considered to be almost the only desulphurisation product accumulating in the headspace during both MA18 and MA14 (Díaz et al., 2010b). In fact, negligible concentrations of both sulphate and thiosulphate were infrequently found in the effluent. Thus, it was estimated that most of the oxygen supplied (60–70%) was consumed in unidentified processes at both 14 and 18 d of HRT. Among them, dissolved sulphide oxidation did not occur, which agreed with the mixing mode of the digester (Díaz et al., 2010b).

4. Conclusion

Oxygen could increase the ability of reactors to handle overloads by forming more stable digestion systems. It was hypothesised that micro-oxygenation prevented a severe imbalance resulting from the hydraulic overload by promoting growth of hydrogenotrophic methanogens, which in turn could favour acetic formation and could help to maintain pH. Furthermore, higher biogas productivity and methane yield were obtained during the period of imbalance, which was related to increased activity rates of acidogens. However, during steady-state operation, oxygen had impact only on the biogas quality. It hardly affected the content of methane and oxygen, while hydrogen sulphide concentration decreased by 90%.

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